

Accretion-Modified Dynamical Friction and Stone Skipping of Black Holes in Ultralight Dark Matter

(Working Title — Taproot Draft)

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Status note. This document exists to establish scope and structure before any writing that could end up in a paper. Nothing in it is checked to submission standard. Every quantitative claim below is either (a) directly reproducible from a script/run in this repository, tagged as such, or (b) marked **[OPEN: ...]** / **[TODO: ...]** as not yet done. Treat this as a lab notebook that happens to be typeset, not a manuscript draft.

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1 Research questions

Two prior, independent results treat the black hole (or test-mass perturber) as having *fixed mass* throughout the simulation:

- 1. Dynamical friction in a uniform ULDM background.** Lancaster, Giovanetti, Mocz, Kahn, Lisanti & Spergel (2019), arXiv:1909.06381, is the foundational analytic + numerical treatment. Wang & Easther (2021), arXiv:2110.03428, extends this to a point mass in a uniform background and then an SMBH embedded in a soliton, establishing baseline friction timescales.
- 2. Stone skipping in a soliton.** Zhang, Wang, Zagorac & Easther (2026), arXiv:2602.11512, shows that a black hole orbiting inside a soliton can undergo non-monotonic, quasi-periodic radial excursions (“stone skipping”), driven by a dipole resonance between the orbit and a soliton oscillation mode, modelled as a forced damped harmonic oscillator.

The proposed new work: repeat both scenarios with the black hole’s mass allowed to *grow via accretion* of the surrounding ULDM, using the sink mechanism already implemented in this codebase (`PyUltraLight2.Particles.Accretion`), and ask whether/how mass growth modifies each phenomenon:

- **Modified dynamical friction:** does a growing $M(t)$ accelerate or otherwise qualitatively change the inspiral, given that dynamical friction forces scale $\propto M^2$ in the classical (Chandrasekhar-type) treatment? Is there a runaway/self-reinforcing regime?
- **Modified stone skipping:** does mass growth detune the dipole resonance (since the resonance condition is mass-ratio dependent), damp the oscillation, or alter its period/amplitude over the run?

[OPEN: These are two separate simulation campaigns (uniform background vs. soliton), not one. Neither has been started yet beyond the accretion machinery validation below.]

[OPEN: Unresolved wrinkle for the dynamical-friction campaign specifically: this project’s own unpublished exploration (uniform background + full self-gravity, *no* accretion) already found runaway gravitational collapse rather than steady dynamical friction — a qualitatively different baseline behaviour from Lancaster et al.’s regime. Needs to be understood/ reconciled (transient-wave-structure regime where their analytics break down? parameter choice?) before accretion is layered on top, or the “modified dynamical friction” framing may need to lead with a different baseline.]

2 Physical setup (as implemented)

Ultralight/fuzzy dark matter obeys the Schrödinger–Poisson system

$$i\hbar \partial_t \psi = \left[-\frac{\hbar^2}{2m} \nabla^2 + m\Phi \right] \psi, \quad \nabla^2 \Phi = 4\pi G m |\psi|^2, \quad (1)$$

evolved pseudospectrally (operator-split kick–drift–kick, FFT Poisson solve) on a periodic grid. A classical point-mass black hole is coupled two-way: it feels Φ (interpolated, RK4-integrated motion) and sources a Plummer-softened potential $\Phi_{TM}(r) = -am/\sqrt{1+a^2r^2}$ added to the field.

Accretion is added as a local imaginary potential (a moving sink):

$$\psi \mapsto \exp[-h V_{\text{sink}}(\mathbf{r}, t)] \psi, \quad V_{\text{sink}}(\mathbf{r}) = A \exp\left(-\frac{|\mathbf{r} - \mathbf{r}_{\text{BH}}|^2}{2R^2}\right), \quad (2)$$

with the removed mass $\Delta M = V_{\text{cell}} \sum (\rho_{\text{before}} - \rho_{\text{after}})$ added to the black hole, and (optionally) the removed momentum $\int \mathbf{j}(\mathbf{r})(1 - e^{-2hV_{\text{sink}}}) dV$, with $\mathbf{j} = \text{Im}(\psi^* \nabla \psi)$, deposited onto its velocity as a rocket-equation update (drag/recoil from the swallowed matter). A, R can be fixed, or dynamically recalibrated every step from the particle’s instantaneous mass and speed to track the classical Bondi–Hoyle–Lyttleton rate,

$$\dot{M}_{\text{BHL}} = \frac{4\pi(GM)^2 \rho}{(v_{\text{rel}}^2 + c_s^2)^{3/2}} \Rightarrow R = \frac{2M}{v_{\text{rel}}^2 + c_s^2}, \quad A = \frac{(v_{\text{rel}}^2 + c_s^2)^{3/2}}{8\sqrt{2\pi} M}. \quad (3)$$

v_{rel} is the black hole’s velocity relative to the *local* ULDM flow (interpolated current $\mathbf{j}/\rho$ at the particle’s position), not its raw box-frame speed — see §4 for why that distinction turned out to matter. Full API in `UserManual.tex` (this repository).

3 Related work and positioning

Fixed-mass dynamics baselines (§??, item list above): Lancaster et al. 2019 (1909.06381); Wang & Easther 2021 (2110.03428); Zhang, Wang, Zagorac & Easther 2026 (2602.11512, “stone skipping”).

Accretion literature — checked directly (live search, not memorised) 2026-09-29, since a first pass at this framing risked an inaccurate “nobody has looked at BH+FDM accretion” claim:

- Cardoso, Ikeda, Vicente & Zilhão (2022), PRD Letters, arXiv:2207.09469, “Parasitic black holes” — full numerical relativity, spherically symmetric, a *static* BH swallowing a boson star. Different regime (GR near-horizon, no moving particle, no BHL rate). Low direct methodological overlap, but closest-sounding title — must cite to preempt “have you seen this?”.
- Chiu, Schive, Yang, Huang & Gaspari (2025), PRL, arXiv:2501.09098 — *static* soliton potential boosting Bondi accretion of a **separate baryonic gas component**; the FDM field itself is never absorbed.
- Ludwig, Mocz & Robles (2026), PRD, arXiv:2607.00349 (submitted Jul 2026) — **closest prior work**. Live, moving BH in a live Schrödinger–Poisson soliton, explicitly benchmarked against $\dot{M}_{\text{BHL}}$. Sink removes mass from a **separate isothermal gas mesh** (FLASH-style cloud-in-cell deposition, strictly mass-conserving), not the ULDM wavefunction; sink radius **fixed** at $2.5\Delta x$ (grid-tied); BHL used only as an external diagnostic target, never fed back into the sink’s own geometry.

Defensible novelty niche (not “first to study this”): (a) direct non-Hermitian absorption of the dark matter field itself, not a gas proxy; (b) sink geometry dynamically slaved to BHL every step from instantaneous (M, v_{rel}) , not fixed-radius-with-external-diagnostic; (c) explicit momentum/recoil deposition from the absorbed field’s local current, not a pure mass leak. **[TODO: Related-work section of any actual manuscript must explicitly position against Ludwig et al. 2607.00349 — omitting it reads as either unaware of very recent literature or dodging it.]**

4 Validation status of the accretion machinery

Everything in this section is directly reproducible from this repository (see `CHANGES.md` for the dated, itemised history; `Analysis/` for the checking scripts).

- **Mass conservation:** exact to $\sim 10^{-12}$ relative (grid mass lost = sink mass gained), checked at every resolution tried so far (32–64, §5).
- **Momentum conservation:** direct check via $\int \text{Im}(\psi^* \nabla \psi) dV + M \mathbf{v}$ at 48^3 . Gravity-only baseline has residual drift $\sim 10^{-3}$ (grid/interpolation noise floor). Sink *without* momentum deposit: drift -0.35 ($\sim 310\times$ the floor — the mass-leak-with-no-recoil problem was real). Sink *with* it: drift $+0.0055$ ($\sim 5\times$ the floor, $64\times$ improvement). **[OPEN: Residual not at machine precision; not yet root-caused (candidates: finite- h discretisation, or consistent with the same noise floor the baseline shows).]**
- **BHL calibration:** 7-point scan (uniform background at rest, particle at fixed velocity, sidesteps a periodic-field phase-closure issue by construction) — measured $\dot{M}$ matched $\dot{M}_{\text{BHL}}$

to within 0.4–1.0% at every point. **[OPEN: This validates the formula’s normalisation only; by construction it cannot distinguish grid-frame speed from relative-to-medium speed, since the two are identical when the background is at rest.]**

- **Relative-velocity correction:** adding `Sink.RelativeVelocity` (interpolated local ULDM velocity subtracted before the BHL calibration, rather than raw box-frame speed) changes accreted mass in `BHAccretionDemo.uldm` by up to +2.3% mid-run, settling to +0.9% — systematic, not noise. Matters because an orbiting BH generically excites bulk motion in the soliton it orbits (the stone- skipping mechanism itself), so the box-frame approximation is not safe in exactly the regime campaign 2 needs.
- **N-body/FDM coupling fixes** (general, not accretion-specific): periodic-boundary force wraparound (inherited bug, fixed); one-substep temporal lag between sink absorption and the Newtonian back-reaction (fixed, ~ 0.1 – 0.13% effect on repeated-passage accretion).

5 Resolution convergence (in progress)

Scan over $N \in \{32, 64, 128, 192, 256\}$ on `BHAccretionDemo.uldm` (fixed `BoxLength=3`, `Duration=1.2`, `Dynamic=ConserveMomentum=RelativeVelocity=true`), tracking final accreted mass and the full `BHMass.npy` curve at each resolution.

Scan complete. All five resolutions finished overnight 2026-09-29/30. Clean, monotonic convergence, successive relative change roughly halving at each step:

N	wall time	final BH mass	rel. change from previous
32	2.5 s	0.751820	—
64	63.6 s	0.740708	−1.478%
128	1846.2 s (≈ 31 min)	0.733825	−0.929%
192	12030.7 s (≈ 200 min)	0.731595	−0.304%
256	57746.0 s (≈ 16.0 h)	0.730552	−0.143%

Practical reading: 128^3 – 192^3 appears to be an adequate resolution floor for this configuration (sub-1% and sub-0.5% of the 256^3 answer respectively), pending the same check in the actual campaign geometries (uniform background / soliton), which may resolve differently.

[OPEN: Wall-time scaling was far steeper than a naive $N^3 \times N^2$ (grid cells $\times$ timesteps) estimate suggested, and got worse with resolution: $64 \rightarrow 128$ was $29\times$ for a $2\times$ bump, $128 \rightarrow 192$ was $6.5\times$ for $1.5\times$, $192 \rightarrow 256$ was $4.8\times$ for $1.33\times$. 256^3 alone took 16 hours on the test machine. Confirmed via repeated ps/log checks during the run that this was steady, non-stalled progress (CPU 165–357%, regular snapshot cadence) rather than a hang — reads as thermal throttling under sustained multi-hour load, not a code bug, but unconfirmed on other hardware. This has real budget implications for the two campaigns below if they need $\gtrsim 192^3$ for extended durations.]

6 Immediate next steps

1. Finish §5; decide a resolution floor for the two campaigns based on it.
2. Resolve the uniform-background collapse wrinkle (§1) before or alongside starting the dynamical-friction campaign.

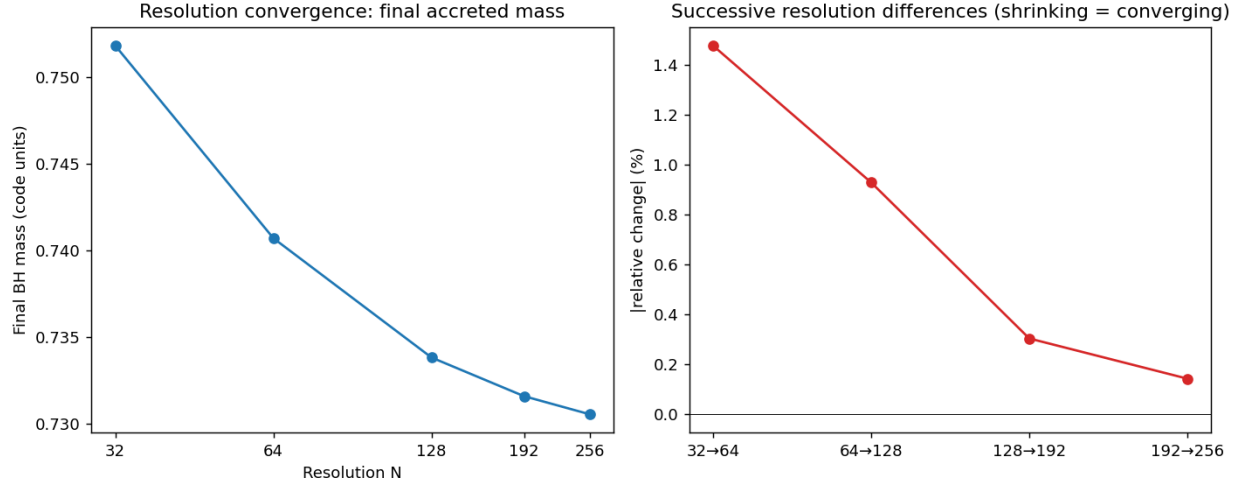


Figure 1: Final accreted mass vs. resolution (left) and successive relative differences (right), all five resolutions. Successive difference shrinks $1.48\% \rightarrow 0.93\% \rightarrow 0.30\% \rightarrow 0.14\%$, roughly halving each step — a clean convergence signature. 192^3 is within 0.3% of the 256^3 value; 128^3 within 0.9% .

3. Scope and run the stone-skipping campaign: reproduce Zhang et al.’s soliton + fixed-mass baseline first (sanity check against their Fig. 1), then re-run with accretion on.
4. `MassChange` and `Sink.Feedback` both write particle mass each step and will clobber each other — needs reconciling before any campaign that might touch `MassChange`.